

SYLLABUS OF MATH 142 BASIC CALCULUS II

2025-2026 SPRING SEMESTER, IZTECH

Instructors:

Inst. Dr. Barış ÇİÇEK bariscicek@iyte.edu.tr

Classroom: Prof. Dr. Erdal SAYGIN Lecture Hall

Course Schedule (Theory): Monday 9:45, 10:45, 11:45

Res. Asst. Dr. Aylin BOZACI SERDAL aylinbozaci@iyte.edu.tr

Classroom: Prof. Dr. Erdal SAYGIN Lecture Hall

Course Schedule (Theory): Monday 13:30, 14:30, 15:30

Res. Asst. Dr. Cihan SAHİLLİOĞULLARI cihansahilliogullari@iyte.edu.tr

Classroom: Prof. Dr. Erdal SAYGIN Lecture Hall

Course Schedule (Theory): Wednesday 9:45, 10:45, 11:45

Res. Asst. Dr. Sinem BENLİ GÖRAL sinembenli@iyte.edu.tr

Classroom: Prof. Dr. Erdal SAYGIN Lecture Hall

Course Schedule (Theory): Wednesday 13:30, 14:30, 15:30

Friday: tutorials and problem solving session

First Course starts at 10:45 and

Second Course starts at 11:45

Microsoft Teams Code: y888xq6

Please sign up the Teams group for following the announcements!

Textbook: Thomas G, M. Weir, J. Hass, Thomas' University Calculus. Early Transcendentals, Pearson Education, in SI units **15th global edition**

Auxiliary Textbook: J. Hass, M. D. Weir, G.B. Thomas, University Calculus. Early Transcendentals, Addison Wesley, Fourth Edition

Additional Sources:

- 1) Ayres F, Mendelson E, Schaum's outline of Calculus, McGraw-Hill, 1990
- 2) Stewart, Calculus, Metric Version, 8th Edition, 2016
- 3) Marsden J, Weinstein A. Calculus Unlimited, Benjamin, 1981
- 4) Dunham W. The Calculus Gallery: Masterpieces from Newton to Lebesgue, 2004
- 5) Silverman R.A. Essential Calculus and Analytic Geometry, Dover, 2003

Office Hour: Students can ask questions at any time during midday at the Tutoring Center of the mathematics department.

ECTS (European Credit Transfer System): 6 ECTS

The total number of study hours a student is supposed to devote per week: 11 hours

Scheme of The Lectures

- Lectures will be face to face.
- Two hours of tutorials per week, on Fridays: Review and problem solving sessions
- Scope: specific section in chapters 6,8,9,13, and 14 in the textbook, **15th edition.** These sections are itemized below.

Grading Policy: All the midterms will be on Friday during the tutorial sessions. The date and place of the final exam will be announced by the secretary of the mathematics department. We can announce the date on the teams group after the secretary post it.

- 2 Midterm exams, 30% each. $30 \times 2 = 60\%$ in total.
- The final exam 40%
- **All the sessions will take the same exams!**
- **All the sessions will be graded together!**

!!! The passing grade for this course is relative. While a score **below 40 will result in an automatic failure**, a score of **40 or above is necessary but not sufficient to pass**. The final passing grade will be set by considering the statistical distribution of grades (e.g., mean, median) in the class.

The dates of the Midterms:

- **First midterm:** 10 April 2026 Friday
- **Second midterm:** 22 May 2026 Friday
- **Final exam:** To be announced
- **Make-up exam:** 1 July 2026 Wednesday

Course materials:

The course materials, including lecture notes, shall not be disseminated. Each student is supposed to assume the personal responsibility of note-taking during lectures.

Attendance:

Attendance is mandatory. Students are expected to adhere to the attendance regulations of the university. Failure to meet these attendance requirements both for the exams and for the lectures is subject to resulting in a grade designation of N/A.

Learning Outcome: A student following the lectures fairly can gain information about:

- integration and integration techniques
- application of definite integrals
- infinite sequences and series including power series
- partial derivatives and multiple integrals.

Ability Outcome: A student following the lectures fairly can improve his/her following capabilities:

- mathematical thinking and problem solving skills
- using and understanding of mathematical symbols and terminology
- improving and assessing innovative thoughts

**MATH 142
BASIC CALCULUS II**

Detailed Syllabus

NOTE: Holiday weeks are excluded!

Week 1-2 : Chapter 6 – Applications of Definite Integrals,

- 6.1 Volumes Using Cross-Sections,
- 6.2 Volumes Using Cylindrical Shells,
- 6.3 Arc Length,
- 6.4 Areas of Surfaces of Revolution

Week 3-5 : Chapter 8 - Techniques of Integration,

- 8.1 Using Basic Integration Formulas,
- 8.2 Integration by Parts,
- 8.3 Trigonometric Integrals,
- 8.4 Trigonometric Substitutions,
- 8.5 Integration of Rational Func. by Partial Fractions,
- 8.8 Improper Integrals

Week 6-9: Chapter 9- Infinite Sequences and Series,

- 9.1 Sequences,
- 9.2 Infinite Series,
- 9.3 The Integral Test
- 9.4 Comparison Tests
- 9.5 The Ratio and Root Test,

- 9.6 Alternating Series, Absolute and Conditional Convergence,
- 9.7 Power Series,
- 9.8 Taylor and Maclaurin Series,
- 9.10 Applications of Taylor Series

Week 10-12: Chapter 13 - Partial Derivatives,

- 13.1 Functions of Several Variables,
- 13.2 Limits and Continuity in Higher Dimensions
- 13.3 Partial Derivatives,
- 13.4 The Chain Rule,
- 13.7 Extreme Values and Saddle Points

Week 13-14: Chapter 14 - Multiple Integrals,

- 14.1 Double and Iterated Integrals over Rectangles,
- 14.2 Double Integrals over General Regions,
- 14.3 Area by Double Integration

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